

Arithmetic Operators		Comparison Operators		Boolean Operators	
Addition	+	Equal to	==	Logical and	and
Subtraction	-	Not equal to	!=	Logical or	or
Multiplication	*	Greater than	>	Logical not	not
Division	/	Less than	<		
Floor division	//	Greater than or equal to	>=		
Modulus	%	Less than or equal to	<=		
Exponent	**				

- **Arithmetic and number types.** Arithmetic operators will return a `float` unless both inputs are integers (`int` type). If both inputs are `int` type, then most arithmetic operators will return an `int`, except for division (`/`) which always returns a `float`.
- **Chaining comparison operators.** Python allows you to combine more than one comparison operator in an expression. So this is a valid expression:

`1 <= 5 == 6 < 10`

- **String comparisons.** `>`, `<`, `>=`, and `<=` compare strings lexicographically. This is like alphabetical order, except capital letters come before lower case letters (so `"B" < "a"`).
- **Precedence rules.** Python will evaluate arithmetic operators first, then comparison operators, then Boolean operators last. Among Boolean operators, `not` is applied first, then `and`, followed by `or` last.

Practice

1. Evaluate the following expressions.

(a) `True or False and False`

(b) `5 <= 7 != 8 > 6`

(c) `not False or True`

(d) `"fox" < "dog" or "fox" == "rabbit" and "fox" < "frog"`

(e) `True and False == True and False`

(f) `3 + 5 * 2 < 15 and not 4 / 2 - 1 == 4`

2. Why does the expression `False == not True` cause an error?

3. Evaluate the following arithmetic expressions.

(a) `17 // 6`

(b) `17 % 6`

(c) `5 + 7 // 2`

Hint: `%` and `//` have the same precedence as multiplication and division.

(d) `-5 % 12`

Hint: Think about a clock, what time is 5 hours before noon?

(e) `-5 // 12`

4. Write a Python function `hours` that inputs a number of minutes, and then prints how long that is in hours and leftover minutes. For example, `hours(135)` should print 2 hours and 15 minutes.